Chapter 12
Electronic Commerce Systems



Accounting Information Systems, 7e
James A. Hall

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Objectives for Chapter 12

- Be acquainted with the topologies that are employed to achieve connectivity across the Internet.
- Possess a conceptual appreciation of the protocols and understand the specific purposes several Internet protocols serve.
- Understand the business benefits associated with Internet commerce and be aware of several Internet business models.
- Be familiar with risks associated with intranet and Internet electronic commerce.
- Understand issues of security, assurance, and trust pertaining to electronic commerce.
- Be familiar with electronic commerce implications for the accounting profession.

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What is E-Commerce?

The electronic processing and transmission of business data

- electronic buying and selling of goods and services
- on-line delivery of digital products
- electronic funds transfer (EFT)
- electronic trading of stocks
- direct consumer marketing
- electronic data interchange (EDI)
- the Internet revolution

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Internet Technologies

- Packet switching
  - messages are divided into small packets
  - each packet of the message takes a different routes
- Virtual private network (VPN)
  - a private network within a public network
- Extranets
  - a password controlled network for private users
- World Wide Web
  - an Internet facility that links users locally and globally
- Internet addresses
  - e-mail address
  - URL address
  - IP address

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Protocol Functions…

- facilitate the physical connection between the network devices.
- synchronize the transfer of data between physical devices.
- provide a basis for error checking and measuring network performance.
- promote compatibility among network devices.
- promote network designs that are flexible, expandable, and cost-effective.

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Internet Protocols

- **Transfer Control Protocol/Internet Protocol (TCP/IP)** - controls how individual packets of data are formatted, transmitted, and received
- **Hypertext Transfer Protocol (HTTP)** - controls web browsers
- **File Transfer Protocol (FTP)** - used to transfer files across the internet
- **Simple Network Mail Protocol (SNMP)** - e-mail
- **Secure Sockets Layer (SSL)** and **Secure Electronic Transmission (SET)** - encryption schemes

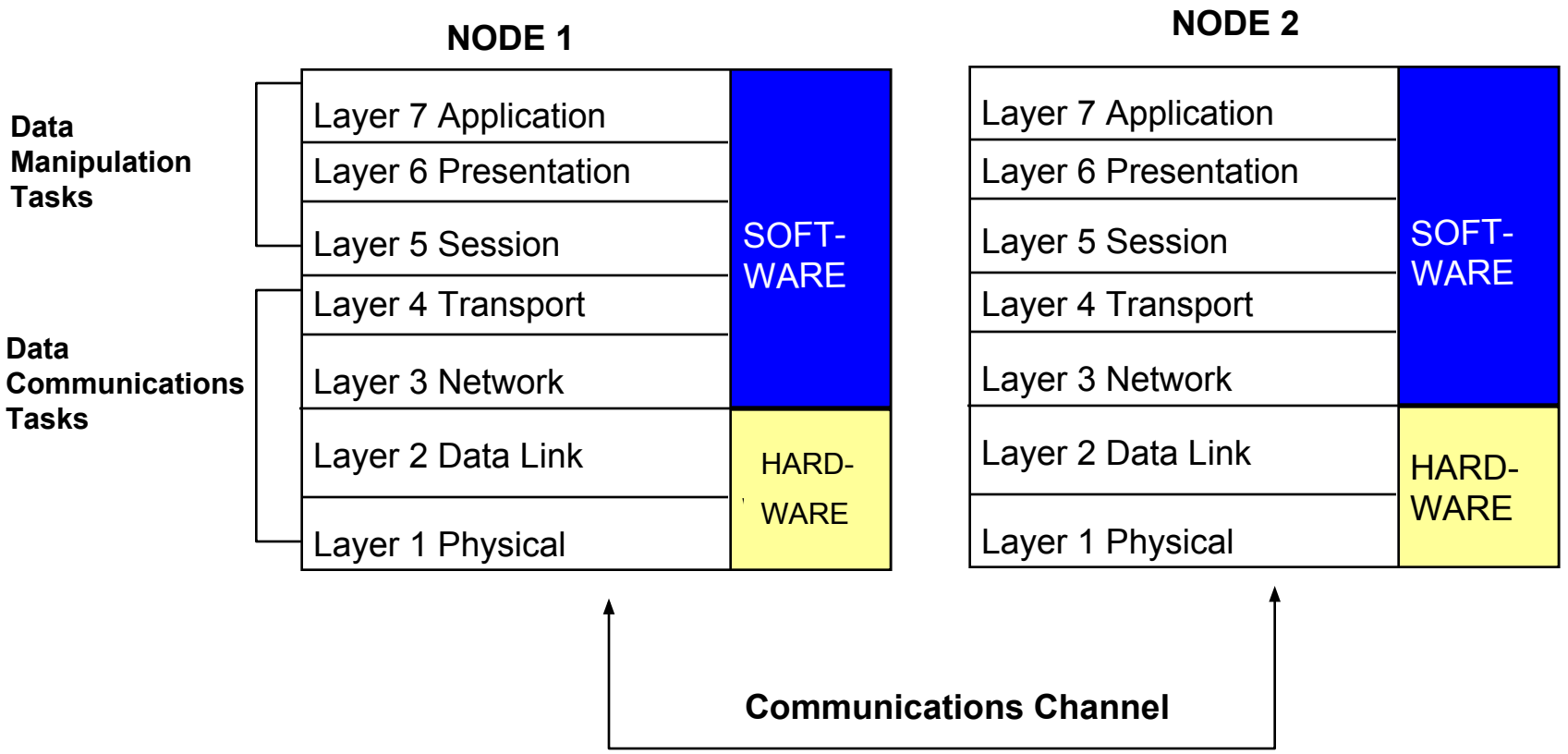
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Open System Interface (OSI)

- The International Standards Organization developed a layered set of protocols called OSI.
- The purpose of OSI is to provide standards by which the products of different manufacturers can interface with one another in a seamless interconnection at the user level.

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The OSI Protocol



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Benefits of E-Commerce

- Access to a worldwide customer and/or supplier base
- Reductions in inventory investment and carrying costs
- Rapid creation of business partnerships to fill emerging market niches
- Reductions in retail prices through lower marketing costs
- Reductions in procurement costs
- Better customer service

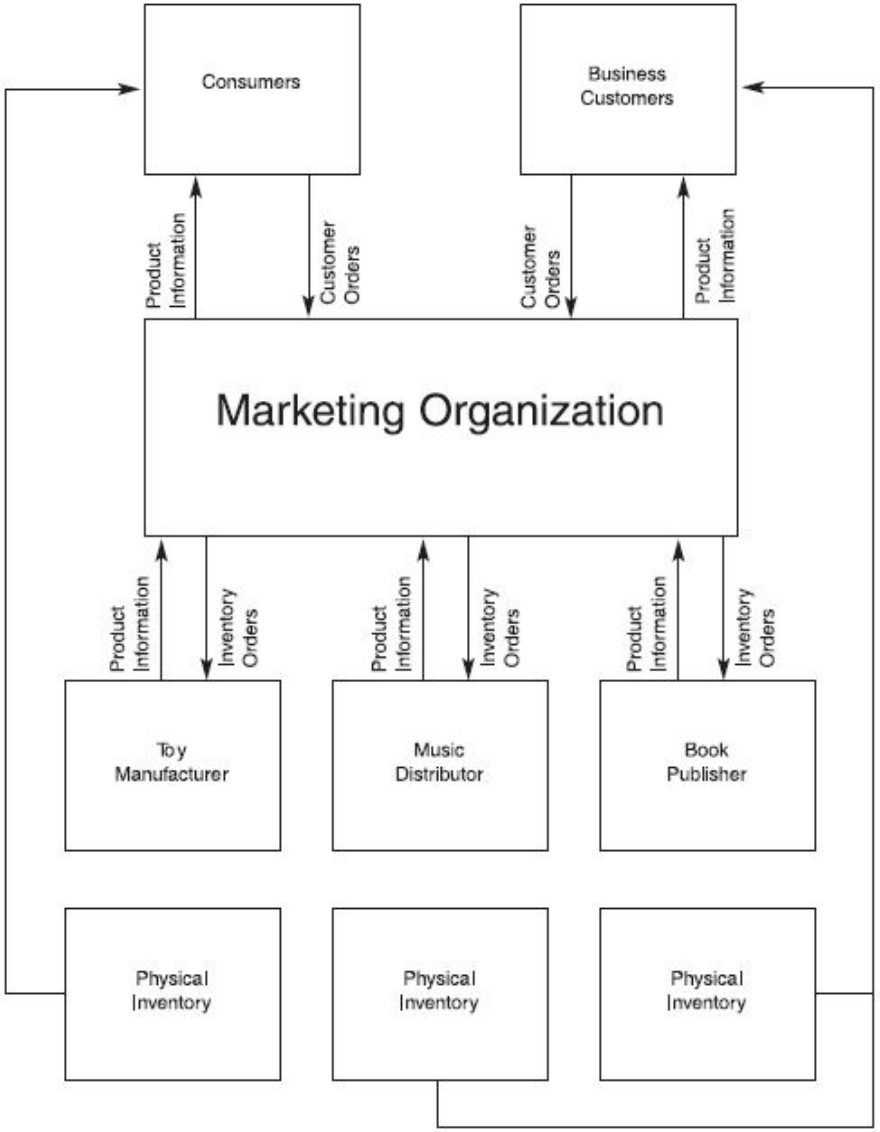
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The Internet Business Model

- *Information level*
  - using the Internet to display and make accessible information about the company, its products, services, and business policies
- *Transaction level*
  - using the Internet to accept orders from customers and/or to place them with their suppliers
- *Distribution level*
  - using the Internet to sell and deliver digital products to customers

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Dynamic Virtual Organization



Perhaps the greatest potential benefit to be derived from e-commerce is the firm's ability to forge dynamic business alliances with other organizations to fill unique market niches as the opportunities arise.

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Areas of General Concern

- *Data Security:* are stored and transmitted data adequately protected?
- *Business Policies:* are policies publicly stated and consistently followed?
- *Privacy*: how confidential are customer and trading partner data?
- *Business Process Integrity*: how accurately, completely, and consistently does the company process its transactions?

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Intranet Risks

- Intercepting network messages
  - sniffing: interception of user IDs, passwords, confidential e-mails, and financial data files
- Accessing corporate databases
  - connections to central databases increase the risk that data will be accessible by employees
- Privileged employees
  - override privileges may allow unauthorized access to mission-critical data
- Reluctance to prosecute
  - fear of negative publicity leads to such reluctance but encourages criminal behavior

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Internet Risks to Consumers

- **How serious is the risk?**
  - National Consumer League: Internet fraud rose by 600% between 1997 and 1998
  - SEC: e-mail complaints alleging fraud rose from 12 per day in 1997 to 200-300 per day in 1999
- **Major areas of concern:**
  - Theft of credit card numbers
  - Theft of passwords
  - Consumer privacy--cookies

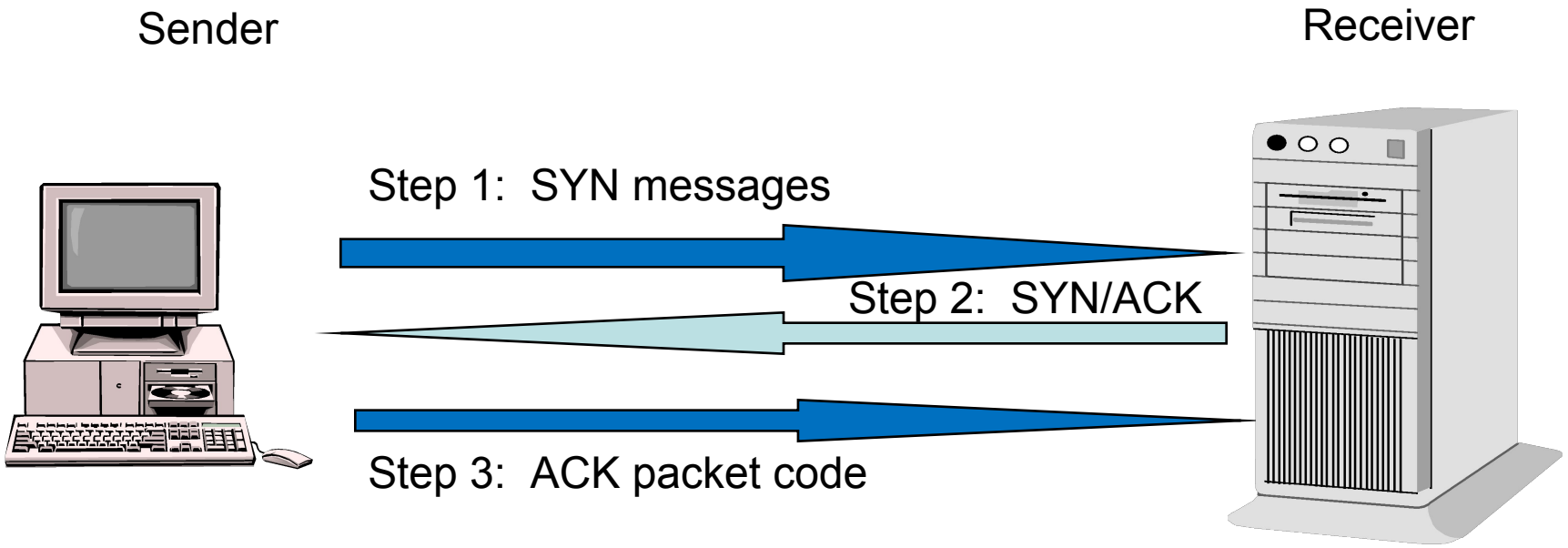
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Internet Risks to Businesses

- *IP spoofing*: masquerading to gain access to a Web server and/or to perpetrate an unlawful act without revealing one's identity
- *Denial of service (DOS) attacks*: assaulting a Web server to prevent it from servicing users
  - particularly devastating to business entities that cannot receive and process business transactions
- *Other malicious programs*: viruses, worms, logic bombs, and Trojan horses pose a threat to both Internet and Intranet users

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SYN Flood DOS Attack



In a DOS Attack, the sender sends hundreds of messages, receives the SYN/ACK packet, but does not response with an ACK packet. This leaves the receiver with clogged transmission ports, and legitimate messages cannot be received.

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Three Common Types of DOS Attacks

- *SYN Flood* – when the three-way handshake needed to establish an Internet connection occurs, the final acknowledgement is not sent by the DOS attacker, thereby tying-up the receiving server while it waits.
- *Smurf* – the DOS attacker uses numerous intermediary computer to flood the target computer with test messages, “pings”.
- *Distributed DOS (DDOS)* – can take the form of Smurf or SYN attacks, but distinguished by the vast number of “zombie” computers hi-jacked to launch the attacks.

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SMURF Attack

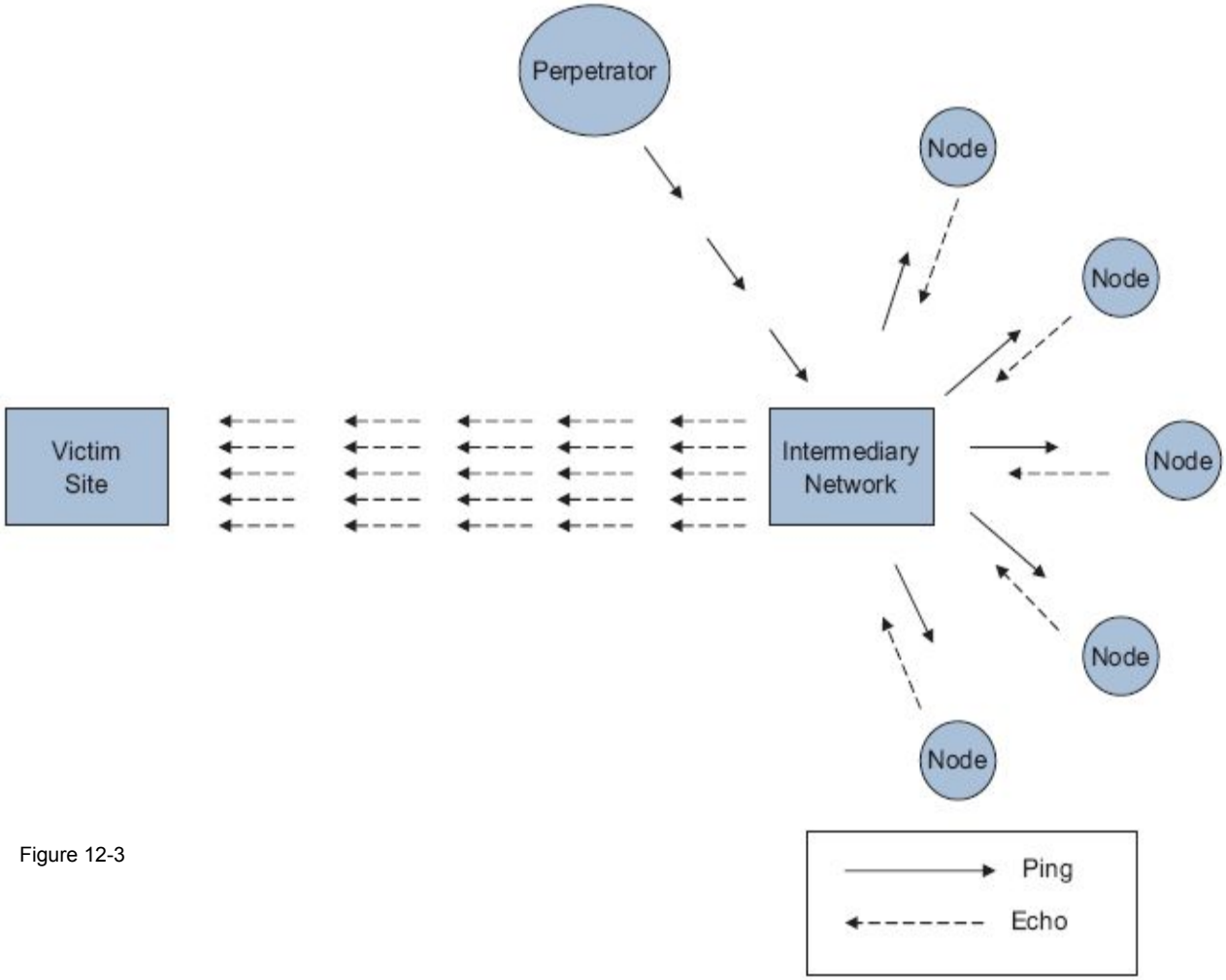


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Distributed Denial of Service Attack

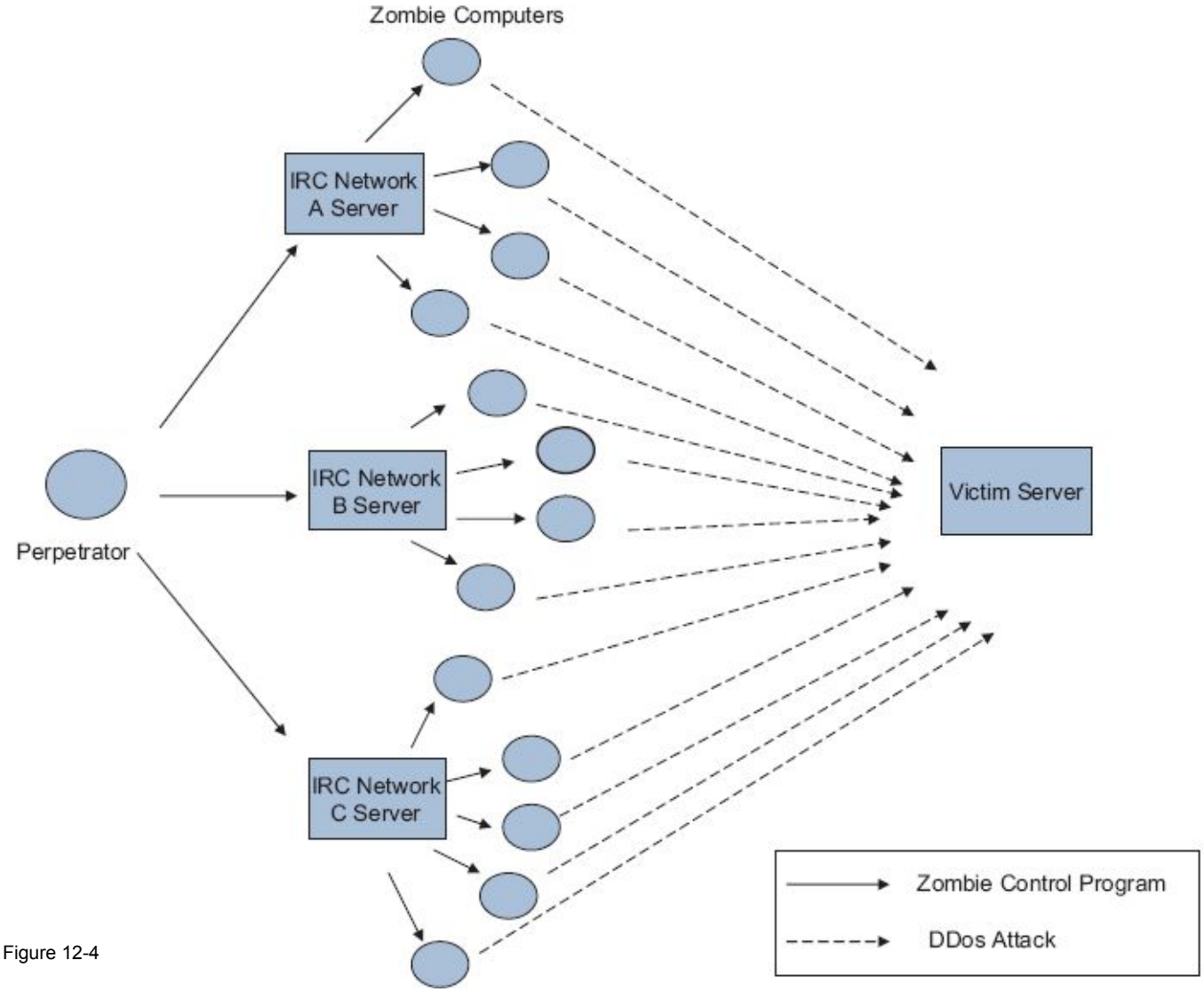


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E-Commerce Security: Data Encryption

- **Encryption** - A computer program transforms a clear message into a coded (ciphertext) form using an algorithm.

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Public Key Encryption

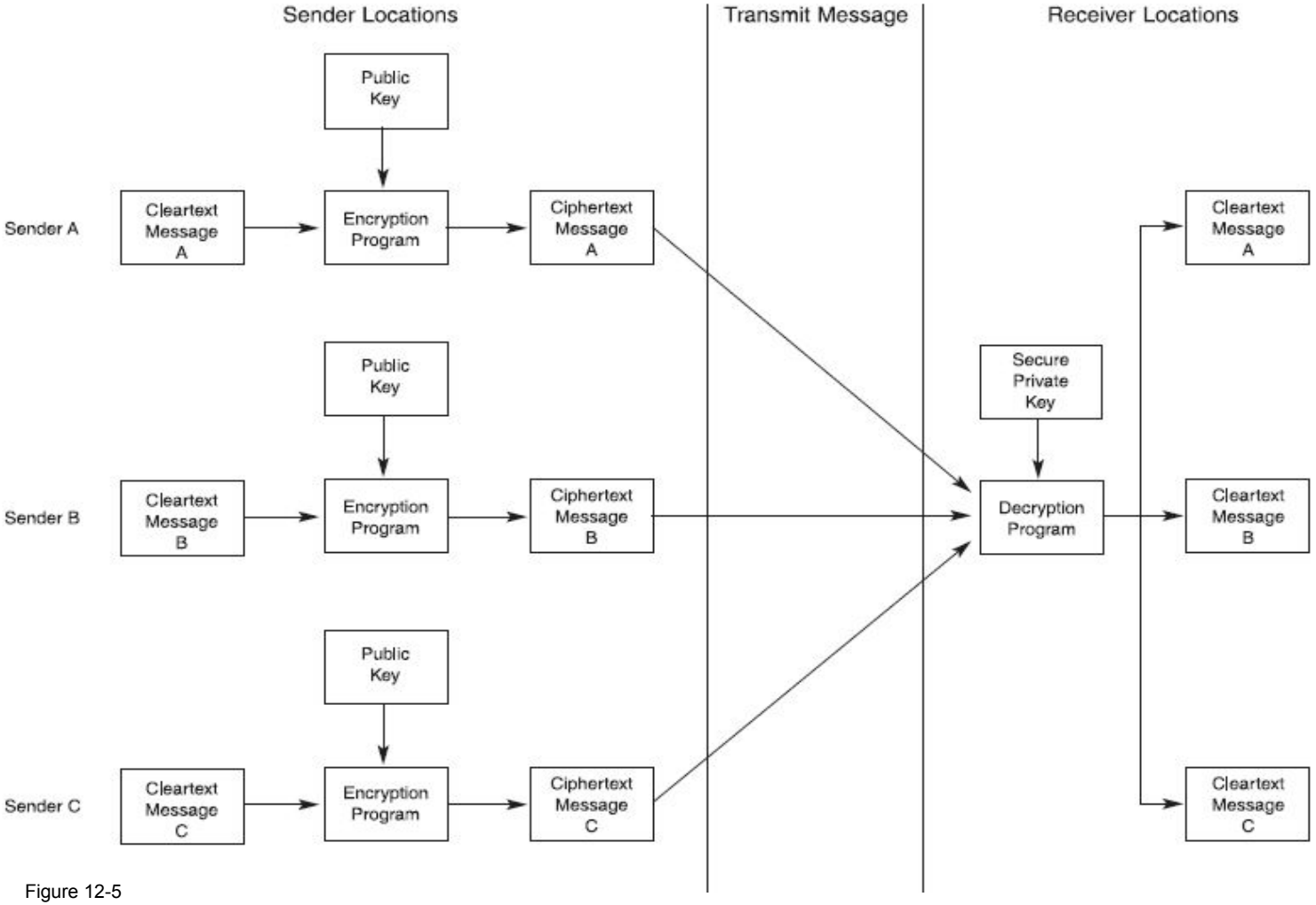


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E-Commerce Security: Digital Authentication

- **Digital signature:** electronic authentication technique that ensures that the transmitted message originated with the authorized sender and that it was not tampered with after the signature was applied
- **Digital certificate:** like an electronic identification card that is used in conjunction with a public key encryption system to verify the authenticity of the message sender

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E-Commerce Security: Firewalls

- **Firewalls:** software and hardware that provide security by channeling all network connections through a control gateway
- **Network level firewalls**
  - low cost/low security access control
  - uses a screening router to its destination
  - does not explicitly authenticate outside users
  - penetrate the system using an IP **spoofing** technique
- **Application level firewalls**
  - high level/high cost customizable network security
  - allows routine services and e-mail to pass through
  - performs sophisticated functions such as logging or user authentication for specific tasks

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Seals of Assurance

- "Trusted" third-party organizations offer *seals of assurance* that businesses can display on their Web site home pages:
  - BBB
  - TRUSTe
  - Veri-Sign, Inc
  - ICSA
  - AICPA/CICA WebTrust
  - AICPA/CICA SysTrust

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Implications for Accounting

- Privacy violation
  - major issues:
    - a stated privacy policy
    - consistent application of stated privacy policies
    - what information is the company capturing
    - sharing or selling of information
    - ability of individuals and businesses to verify and update information captured about them
  - 1995 Safe Harbor Agreement
    - establishes standards for information transmittal between US and European companies

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Implications for Accounting

- Continuous auditing
  - auditors review transactions at frequent intervals or as they occur
  - **intelligent control agents:** heuristics that search electronic transactions for anomalies
- Electronic audit trails
  - electronic transactions generated without human intervention
  - no paper audit trail

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Implications for Accounting

- Confidentiality of data
  - open system designs allow mission-critical information to be at the risk to intruders
- Authentication
  - in e-commerce systems, determining the identity of the customer is not a simple task
- Nonrepudiation
  - repudiation can lead to uncollected revenues or legal action
  - use digital signatures and digital certificates

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Implications for Accounting

- Data integrity
  - determine whether data has been intercepted and altered
- Access controls
  - prevent unauthorized access to data
- Changing legal environment
  - provide client with estimate of legal exposure

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Appendix

Intra-Organizational Electronic Commerce

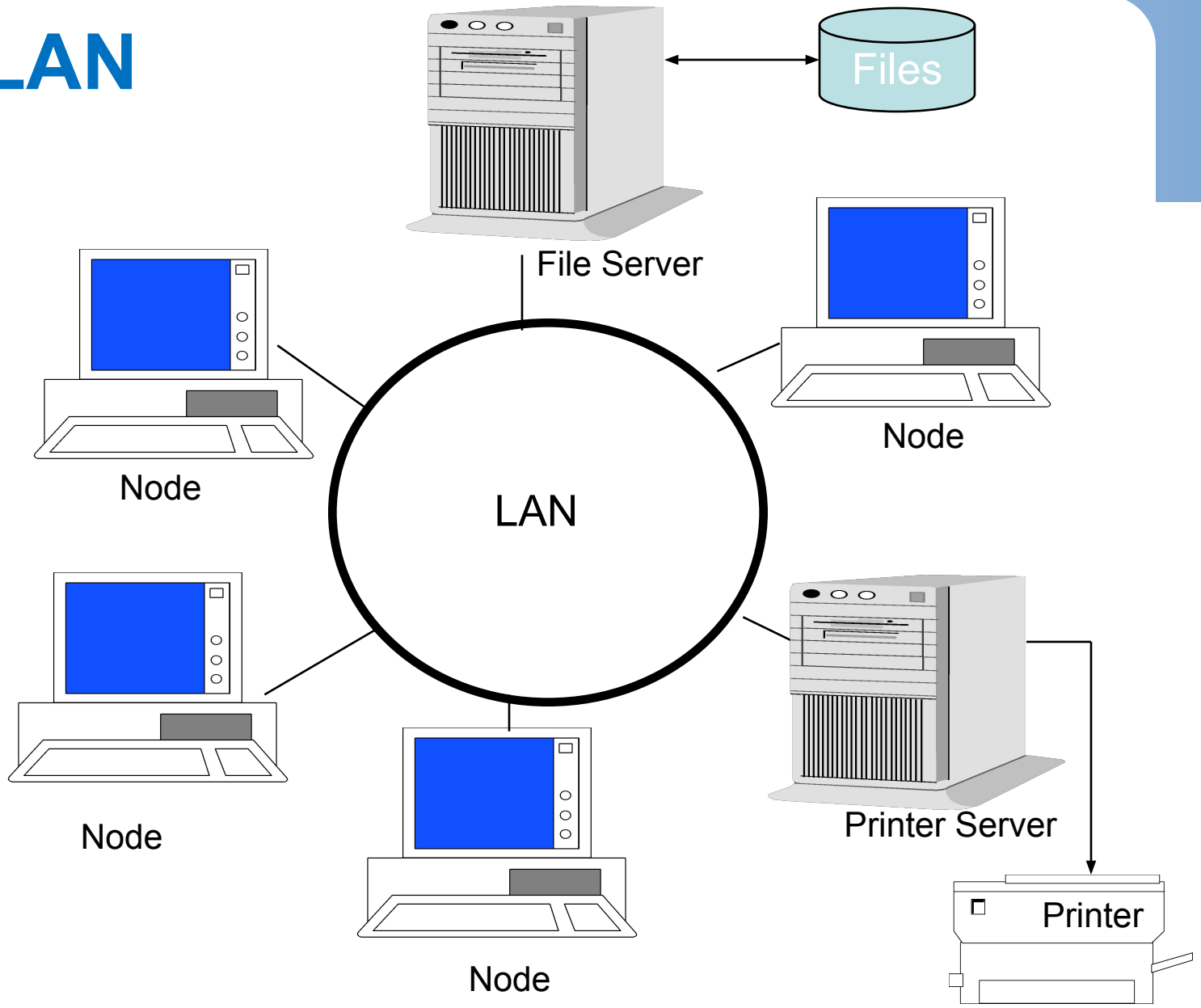
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Local Area Networks (LAN)

- A **federation of computers located close together** (on the same floor or in the same building) **linked together to share data and hardware**
- The physical connection of workstations to the LAN is achieved through a **network interface card** (NIC) which fits into a PC's expansion slot and contains the circuitry necessary for inter-node communications.
- A **server** is used to store the network operating system, application programs, and data to be shared.

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LAN



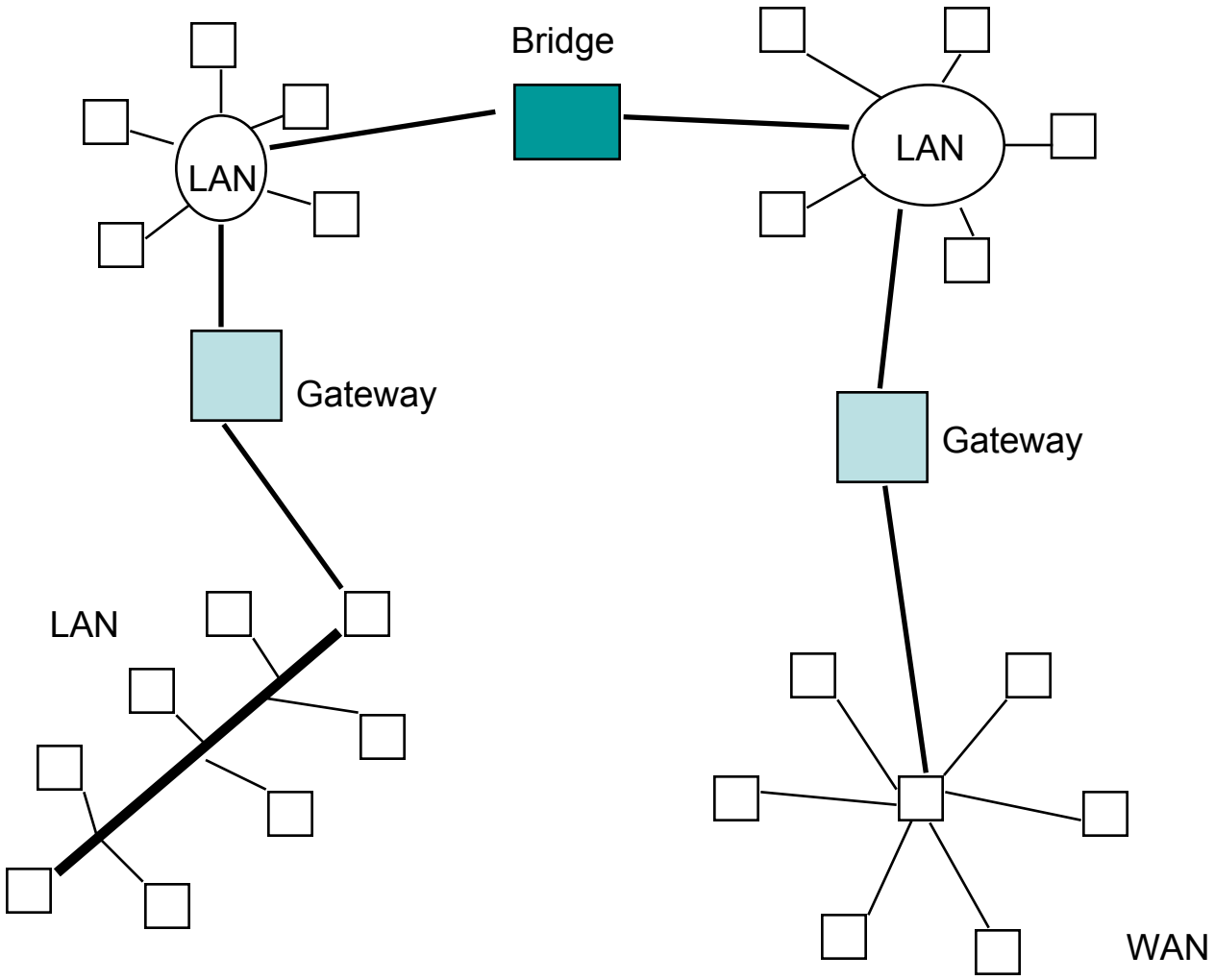
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Wide Area Network (WAN)

- A WAN is a network that is dispersed over a wider geographic area than a LAN. It typically requires the use of:
  - **gateways** to connect different types of LANs
  - **bridges** to connect same-type LANs
- WANs may use common carrier facilities, such as telephone lines, or they may use a **Value Added Network (VAN)**.

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WAN



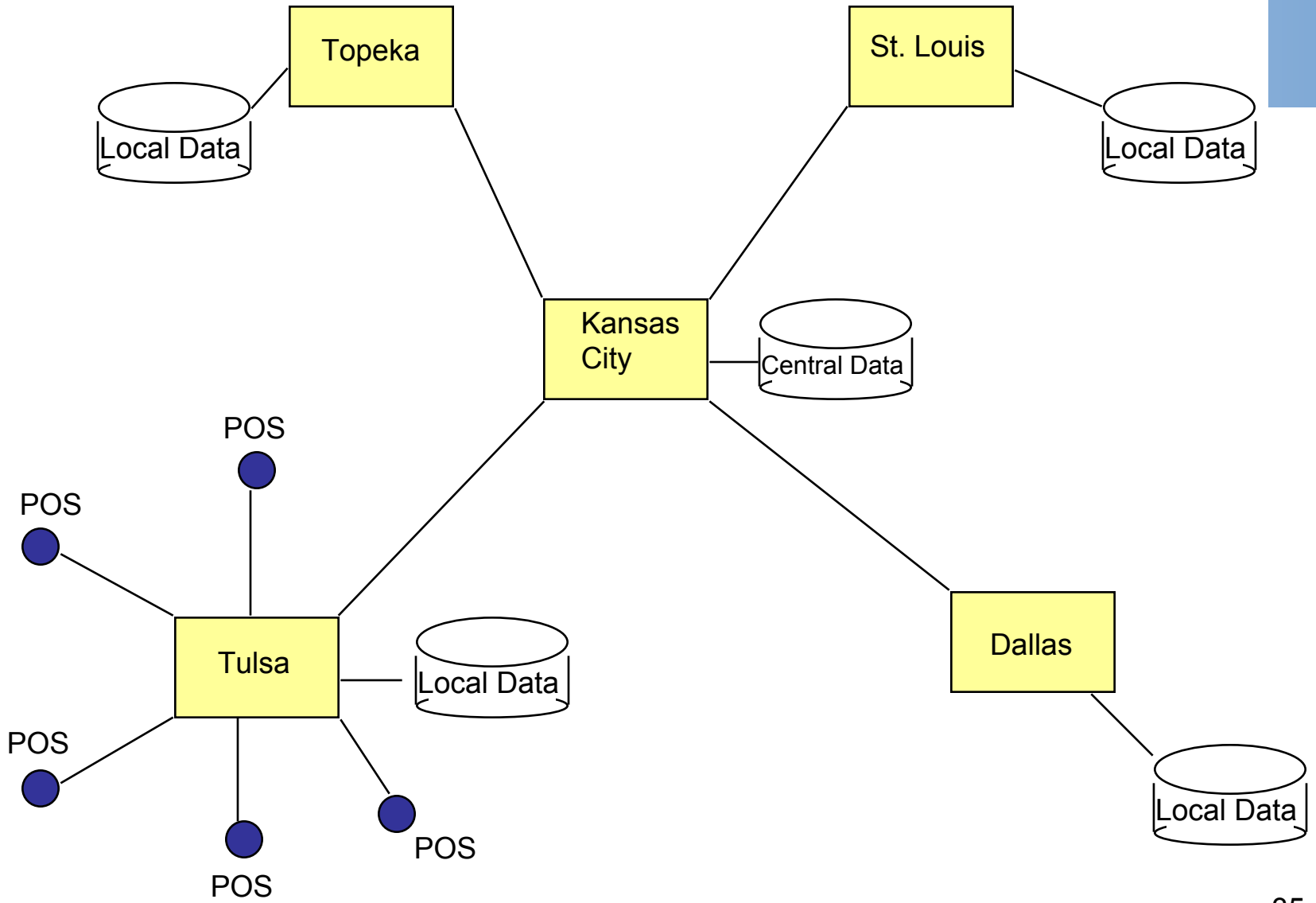
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Star Topology

- A network of IPUs with a large central computer (the host)
- The host computer has direct connections to smaller computers, typically desktop or laptop PCs.
- This topology is popular for mainframe computing.
- All communications must go through the host computer, except for local computing.

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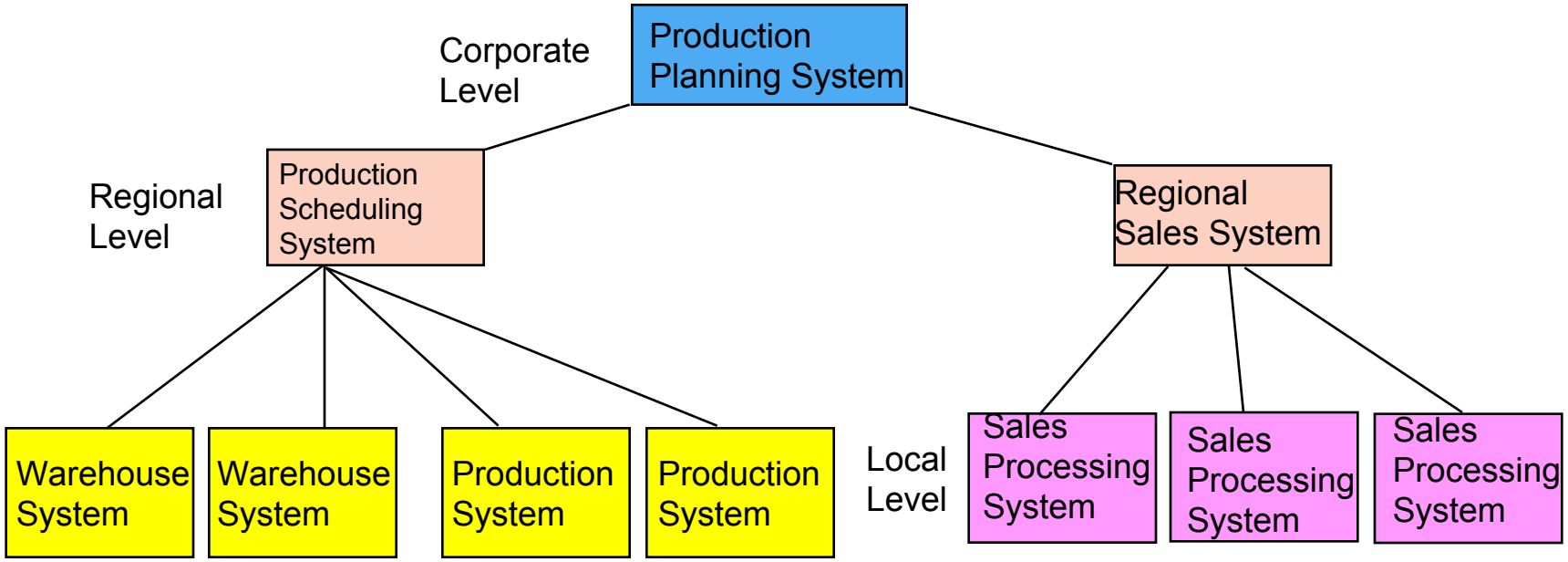
Star Network



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Hierarchical Topology

- A host computer is connected to several levels of subordinate smaller computers in a **master-slave** relationship.



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Ring Topology

- This configuration eliminates the central site. All nodes in this configuration are of equal status (peers).
- Responsibility for managing communications is distributed among the nodes.
- Common resources that are shared by all nodes can be centralized and managed by a file server that is also a node.

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Ring Topology

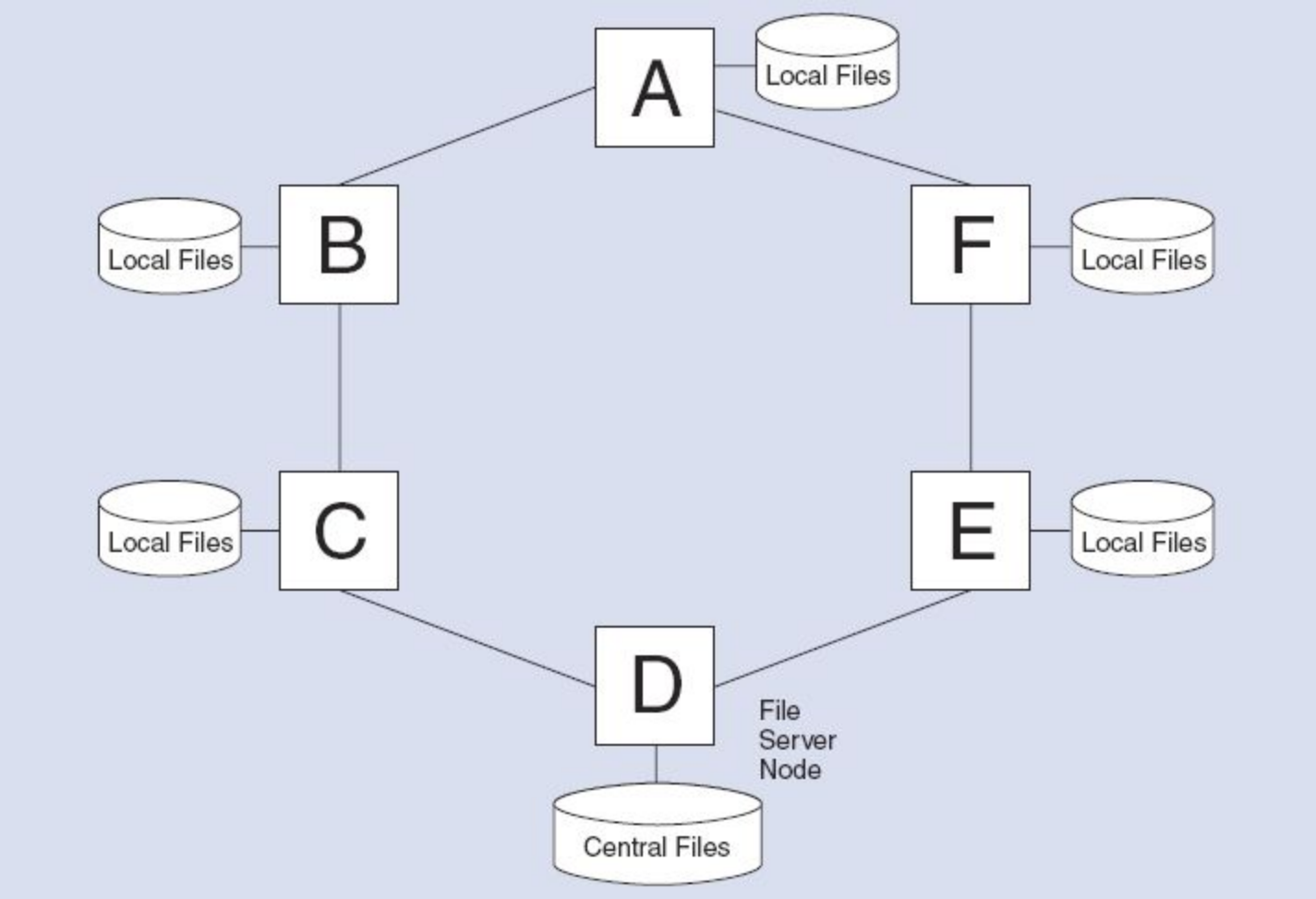


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Bus Topology

- The nodes are all connected to a common cable - the bus.
- Communications and file transfers between workstations are controlled by a server.
- It is generally less costly to install than a ring topology.

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Bus Topology

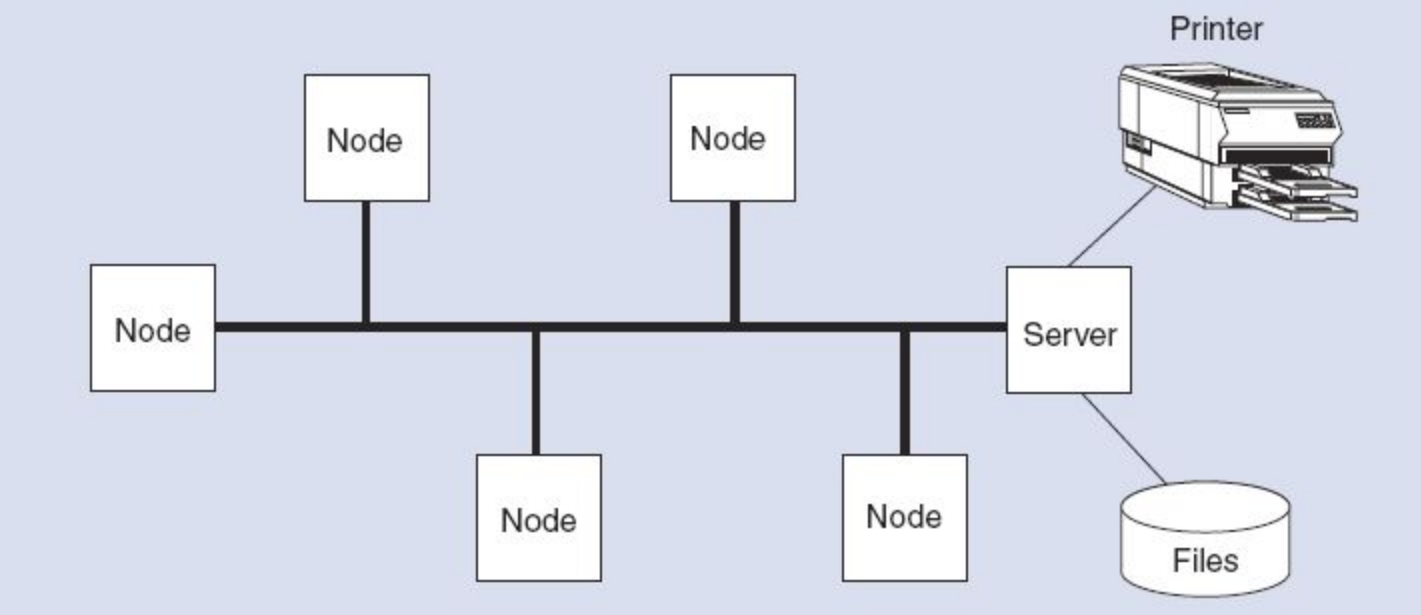


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Client-Server Topology

- This configuration distributes the processing between the user's (client's) computer and the central file server.
- Both types of computers are part of the network, but each is assigned functions that it best performs.
- This approach reduces data communications traffic, thus reducing queues and increasing response time.

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Client-Server Topology

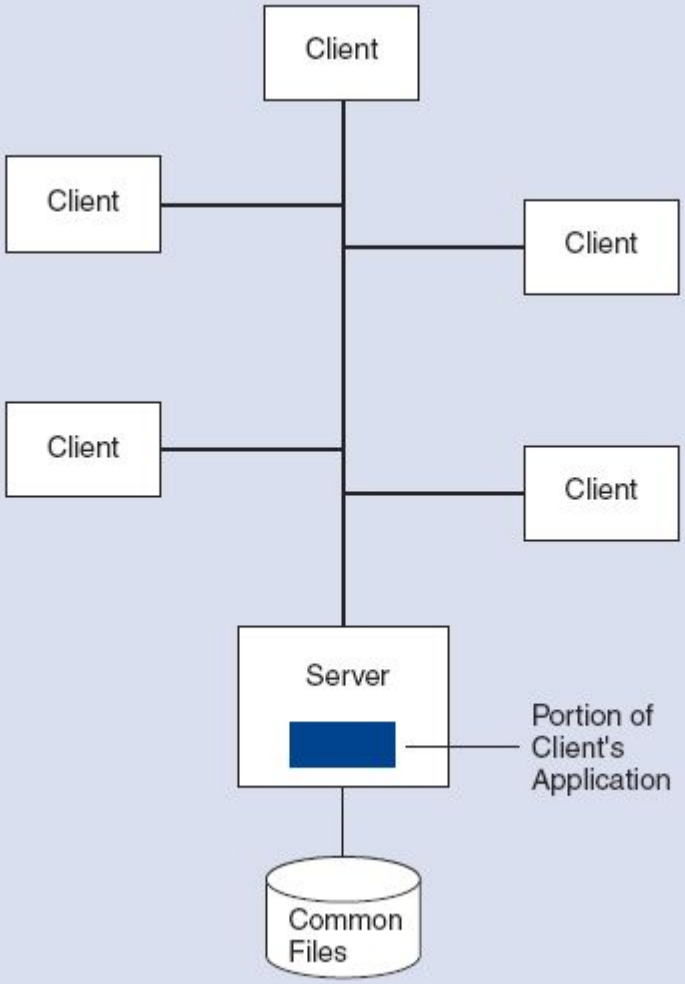


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Network Control Objectives

- **establish a communications session** between the sender and the receiver
- **manage the flow of data** across the network
- **detect errors** in data caused by line failure or signal degeneration
- **detect and resolve data collisions** between competing nodes

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Polling Method of Controlling Data Collisions

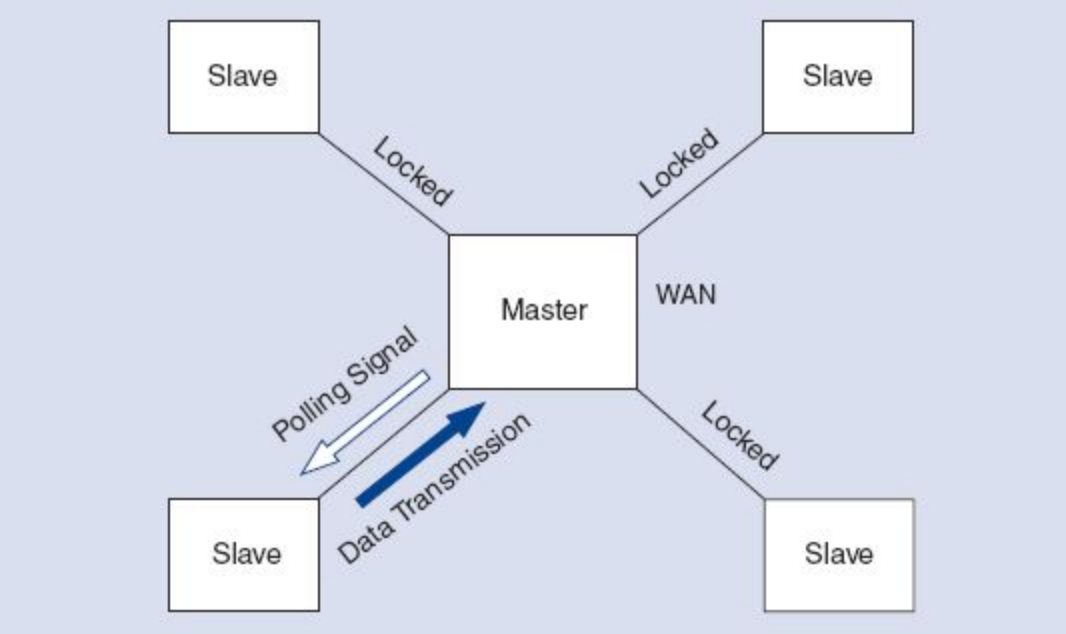


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Token-Passing Approach to Controlling Data Collision

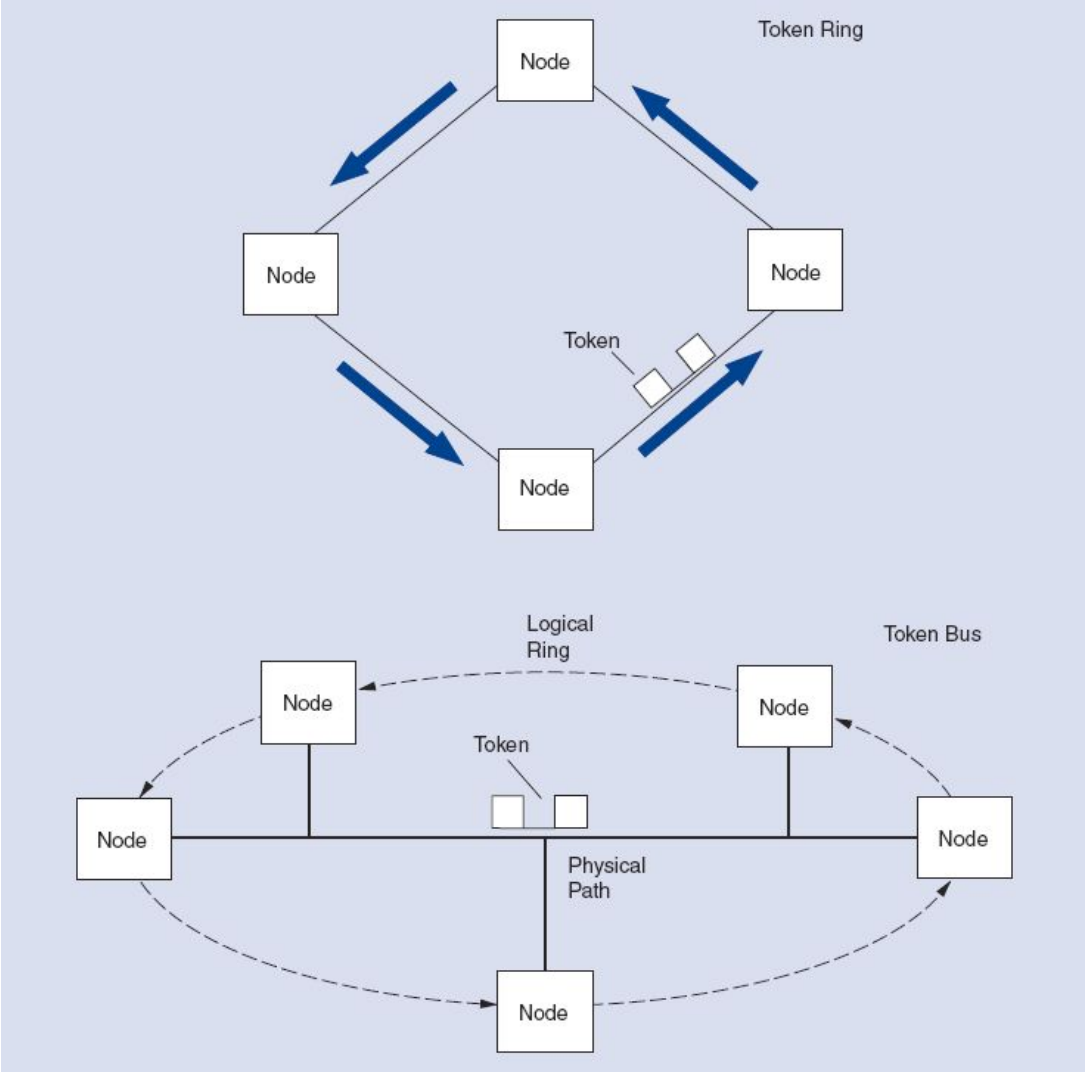


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Carrier Sensing

- ***A random access technique that detects collisions when they occur***
- This technique is widely used--found on *Ethernets.*
- The node wishing to transmit listens to the line to determine if in use. If it is, it waits a pre-specified time to transmit.
- Collisions occur when nodes listen, hear no transmissions, and then simultaneously transmit. Data collides and the nodes are instructed to hang up and try again.
- Disadvantage: The line may not be used optimally when multiple nodes are trying to transmit simultaneously.

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What is Electronic Data Interchange (EDI)?

- The **exchange of business transaction information:**
  - between companies
  - in a standard format (ANSI X.12 or EDIFACT)
  - via a computerized information system
- In "**pure**" EDI systems, *human involvements is not necessary* to approve transactions.

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Communications Links

- Companies may have internal EDI translation/communication software and hardware.

OR

- They may subscribe to VANs to perform this function without having to invest in personnel, software, and hardware.

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Overview of EDI

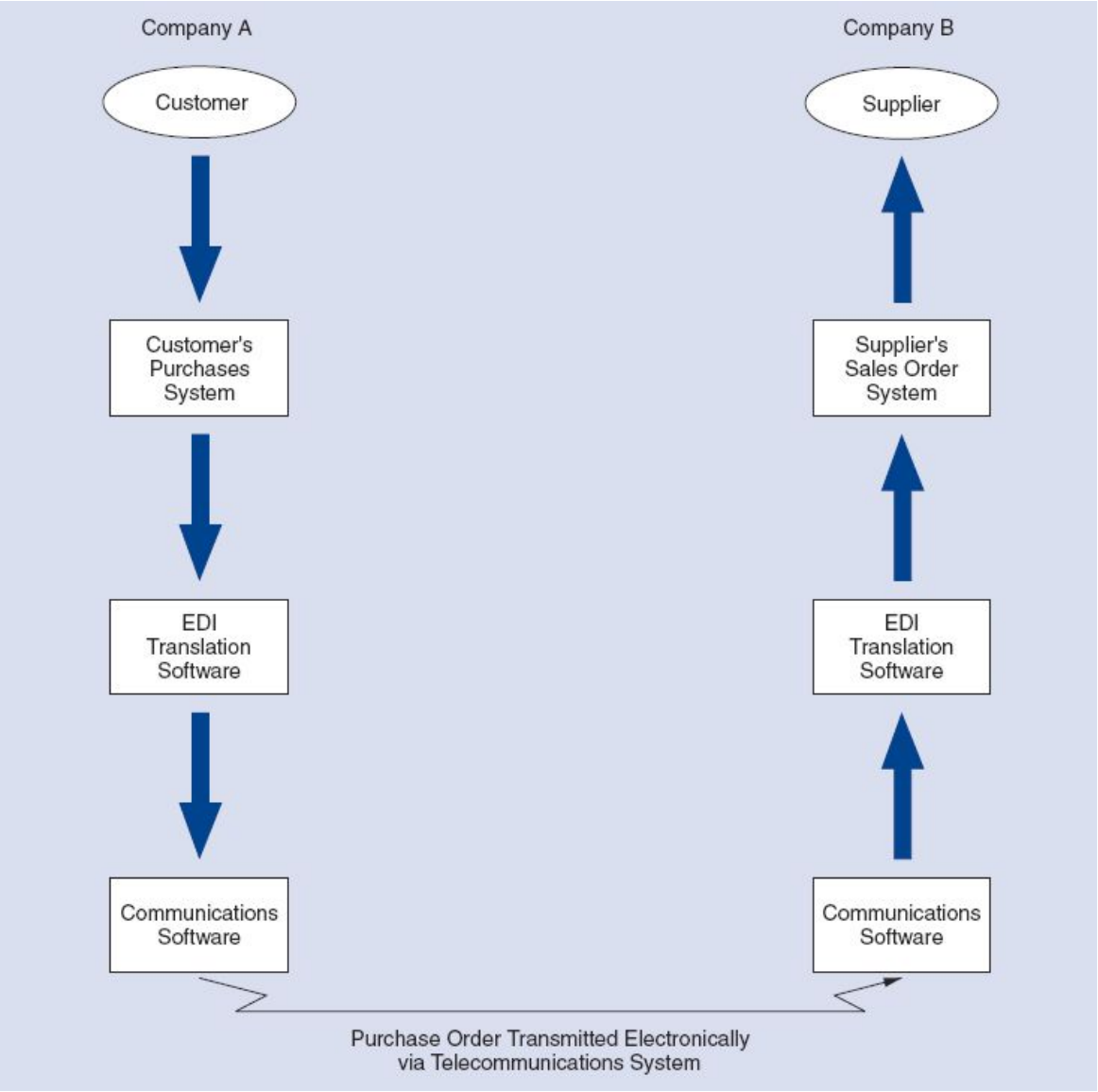


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Advantages of EDI

- Reduction or elimination of data entry
- Reduction of errors
- Reduction of paper
- Reduction of paper processing and postage
- Reduction of inventories (via JIT systems)

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